

Solution Requirements

Requirement Analysis Phase — Phase II

Project Name	Automated Network Request Management in ServiceNow
Phase	Phase 2: Requirement Analysis
Author	Salina Sriraj
Team ID	SALINASRIRAJ-NRM-01
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1. Functional Requirements (FR)

Functional requirements define the specific behaviors, actions, and technical features that the Automated Network Request Management System must execute to fulfill its purpose.

FR No.	Requirement (Epic)	Sub-Requirements & Details
FR1	User Authentication & Role Management	Authenticate all users via the ServiceNow sys_user table. Assign explicit roles: sn_network_requester (IT staff), sn_network_approver (managers/security), sn_network_engineer (fulfillment operators). Enforce ACLs to restrict data visibility per role.
FR2	Service Catalog & Intake Management	Provide 8 distinct, searchable network catalog items (Network Access, Firewall Changes, VPN Access, VLAN Config, IP Allocation, Bandwidth Upgrade, DNS/DHCP Change, Wireless Access). Render dynamic form fields using UI Policies. Apply Client Script regex validation for IP formats, CIDR blocks, and port numbers.
FR3	Multi-Stage Approval Engine	Automatically route high-risk items (firewall, VLAN) to the Network Security Team after Line Manager sign-off. Provide explicit Approve/Reject action buttons in the workspace and email notifications. Enforce mandatory approver comments on rejection. Route low-risk items directly to fulfillment after manager approval.
FR4	Queue Management & Task Assignment	Auto-generate a Catalog Task (sc_task) upon final approval sign-off. Route tasks via Assignment Rules to the correct Network Operations Group or Security Operations queue. Prevent manual task reassignment across unrelated groups.

FR5	Workflow & Notification Automation	Bind all catalog items to Flow Designer workflows. Fire automated email notifications for: Submission Confirmation, Pending Approval Requests, Approval/Rejection Decisions, Task Assignment, and Final Fulfillment Closure.
FR6	SLA Tracking & Escalation	Trigger an SLA time clock upon fulfillment task generation. Execute automated escalation notifications to team leads at 50%, 75%, and 100% of the target resolution window breach threshold.

2. Non-Functional Requirements (NFR)

Non-functional requirements define the operational qualities and constraints governing how the system performs, independent of its specific behaviors.

NFR No.	Requirement	Description
NFR1	Usability	The application exposes a clean, user-centric front-end via the ServiceNow Service Portal, allowing non-technical users to file network requests with the guidance of contextual tooltips and inline help text.
NFR2	Security & Access Control	Data access is governed strictly via Access Control Lists (ACLs) and roles. Requesters can view only their own submissions. Network Engineers can view and edit only records assigned to their specific group queue.
NFR3	Reliability & Auditability	Every approval decision, modification, comment, and task change must write an immutable, date-stamped audit entry into system tables. The database layer must reliably preserve complete transaction histories.
NFR4	Performance	Portal catalog form page-load times must be under 2 seconds. Background business rules, data validations, workflow triggers, and notification dispatches must process with minimal latency.
NFR5	Availability	The system must maintain 99.9% uptime compliance, matching the active operational baseline availability parameters of the enterprise's host ServiceNow cloud instance.
NFR6	Scalability	The schema design (catalog items, Flow structures, variable sets) must scale to accommodate concurrent request volume spikes and support a growing portfolio of integrated infrastructure items without requiring code redesign.